# MCMC Part 2

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# Overview

Today, we cover:

- Gibb's sampling
- Hamiltonian MC (briefly)

Resources:

- Peng Advanced Stat Computing Chapter 7
- Book chapter on Hamiltonian MC

# Single component Metropolis-Hastings

The standard Metropolis algorithm updates the entire parameter vector at once with a single step.

- Hence, a $p$-dimensional parameter vector must have a $p$-dimensional proposal distribution $q$.
- However, it is sometimes simpler to update individual components of the parameter vector one at a time, in an algorithm known as single component Metropolis-Hastings (SCMH).

Let $x^t = (x_1^t, x_2^t, \dots, x_p^t)$ be a $p$-dimensional vector of parameter values at iteration $t$.

# Single component M-H

The SCMH algorithm updates each parameter in $x$ one at a time. All that is needed to update the $i$th component of $x^t$ is a proposal distribution $q(x_i^*|x_i^t, x_{-i}^t)$. SCMH updates the $i$th component via the following steps:

(1) Sample $x_i^* \sim q(x_i^*|x_i^t, x_{-i}^t)$ as a proposal for component $i$.
(2) Let

$$\alpha(x_i^*|x^t) = \min\left\{\frac{\pi(x_i^*|x_{-i}^t)q(x_i^t|x_i^*, x_{-i}^t)}{\pi(x_i^t|x_{-i}^t)q(x_i^*|x_i^t, x_{-i}^t)}, 1\right\}$$

(3) Accept $x_i^*$ with probability $\alpha(x_i^*|x^t)$
(4) Repeat steps 1-3 $p$ times until each component is updated

# Gibb's sampling

- Special case of single component M-H where you always accept the candidate
- For sampling from multivariate distributions (2 or more parameters)
  - Relies on ability to calculate full conditionals

Useful when:

- Sampling from $\pi(\theta_1, \theta_2, \theta_3)$ is hard
- Sampling from $\pi(\theta_1|\theta_2, \theta_3)$, $\pi(\theta_2|\theta_1, \theta_3)$ , $\pi(\theta_3|\theta_1, \theta_2)$ is easy

# Gibb's sampling algorithm

Steps for each iteration:

1. Sample $\theta_1^{n+1} \sim \pi(\theta_1 | \theta_2^n, \theta_3^n)$
2. Sample $\theta_2^{n+1} \sim \pi(\theta_2 | \theta_1^{n+1}, \theta_3^n)$
3. Sample $\theta_3^{n+1} \sim \pi(\theta_3 | \theta_1^{n+1}, \theta_2^{n+1})$

Do this for the desired number of samples, discard burn-in, and voila! You now have a sample from the joint distribution $\pi(\theta_1, \theta_2, \theta_3)$.

# Example: Bivariate normal distribution

Suppose we want to simulate $x = (x_1, x_2)$ from a bivariate Normal distribution with mean $\mu = (\mu_1, \mu_2)$ and covariance matrix

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \rho \\ \rho & \sigma_2^2 \end{pmatrix}$$

we can make use of Gibbs sampling to simulate a Markov chain that will converge to a bivariate Normal.

# Bivariate normal distribution

Goal is to simulate from joint distribution $\pi(x_1, x_2|\mu, \Sigma)$.

Complete conditionals are:

- $\pi(x_1|x_2, \mu, \Sigma) = N(\mu_1 + \rho\frac{\sigma_1}{\sigma_2}(x_2 - \mu_2), \sigma_1^2(1-\rho))$
- $\pi(x_2|x_1, \mu, \Sigma) = N(\mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x_1 - \mu_1), \sigma_2^2(1-\rho))$

So, we sample

1. $x_1^{t+1} \sim N(\mu_1 + \rho\frac{\sigma_1}{\sigma_2}(x_2^t - \mu_2), \sigma_1^2(1-\rho))$
2. $x_2^{t+1} \sim N(\mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x_1^{t+1} - \mu_1), \sigma_2^2(1-\rho))$

# Bivariate normal distribution

```
gibbs_bivariate = function(x0, n = 1000,
                           mu = c(1, 5), rho = 0.5,
                           sigma1 = 1, sigma2 = 2){
  x1 = x0[1]
  x2 = x0[2]
  mu1 = mu[1]
  mu2 = mu[2]

  x = matrix(NA, nrow = n, ncol = 2)
  for(i in 1:n){
    x[i,] = c(x1,x2)

    x1 = rnorm(1, mean = mu1 + rho * sigma1/sigma2 * (x2-mu2),
               sd = sigma1 * sqrt(1-rho))
    x2 = rnorm(1, mean = mu2 + rho * sigma2/sigma1 * (x1-mu1),
               sd = sigma2 * sqrt(1-rho))
  }
  colnames(x) = c("x1", "x2")
  return(x)
}
```

# Bivariate normal distribution

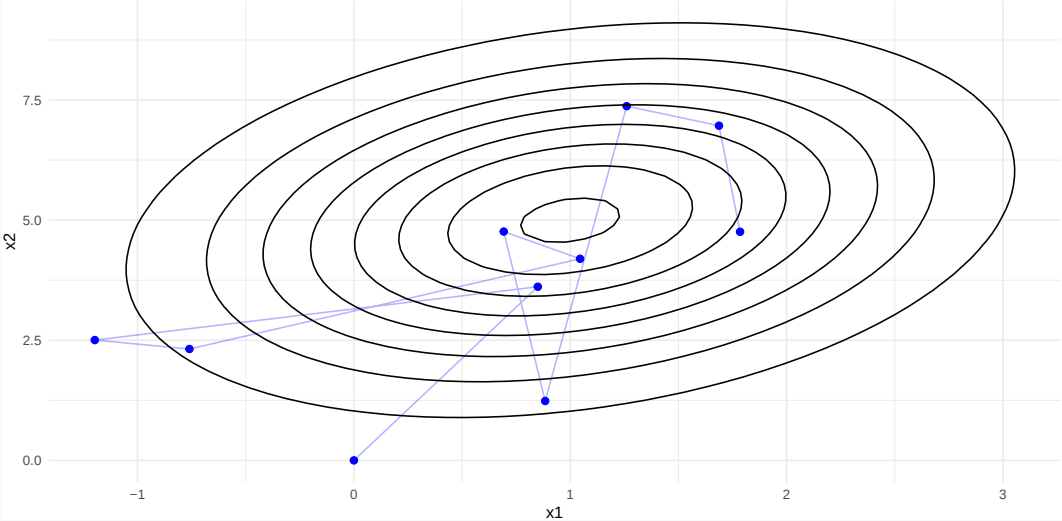
```r
set.seed(21433)
gibbs_x = gibbs_bivariate(c(0,0), n = 10000)

colMeans(gibbs_x[1:100,])
```

```
       x1        x2
0.7881165 4.6632079
```

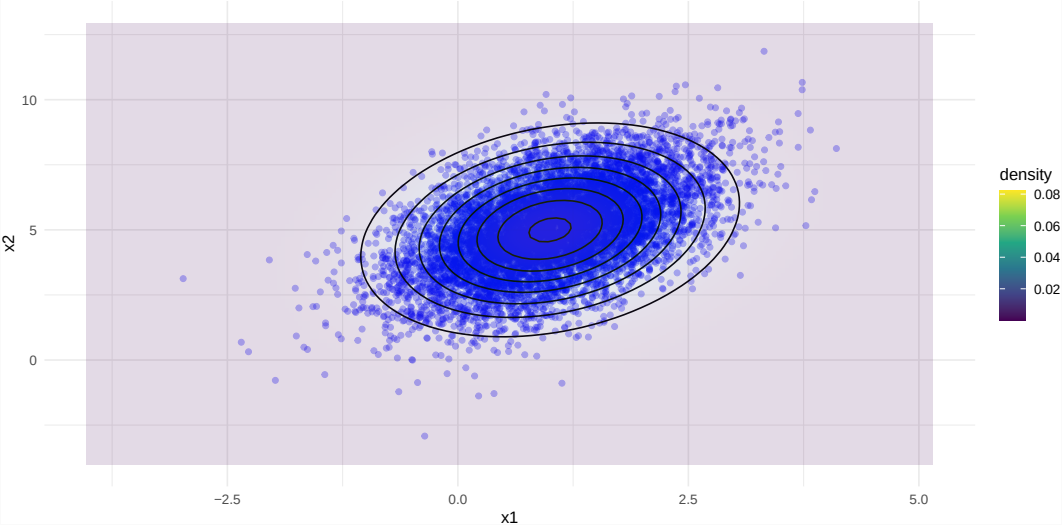
```r
colMeans(gibbs_x[101:1000,])
```

```
       x1        x2
0.9977687 4.9546973
```

# Bivariate normal distribution



# Bivariate normal distribution



# Bivariate normal distribution

Lab- try out the Gibbs bivariate normal sampler when $\rho = 0.05$ and $\rho = 0.95$. Compare results, especially effective sample size and autocorrelation plots.

# Proof: Gibbs sampling as M-H

Gibbs is a special case of single component M-H that always accepts the proposal. Define the following:

- $\pi(\theta_1, \theta_2)$: joint distribution we want to sample from
- $q(\theta_1^* | \theta_1^t, \theta_2^t)$: proposal distribution for new value of $\theta_1$
- Acceptance ratio:

$$\alpha(\theta_1^* | \theta^t) = \min \left\{ \frac{\pi(\theta_1^*, \theta_2^t) q(\theta_1^t | \theta_1^*, \theta_2^t)}{\pi(\theta_1^t, \theta_2^t) q(\theta_1^* | \theta_1^t, \theta_2^t)}, 1 \right\}$$

- Choose full conditional as proposal, i.e.:

$$q(\theta_1^* | \theta_1^t, \theta_2^t) = \pi(\theta_1^* | \theta_2^t)$$

# Proof: Gibbs sampling as M-H

# Bayesian linear regression

$$y = X\beta + \epsilon; \epsilon \sim N(0, \sigma_y^2 I_n)$$

- Need to make (prior) distributional assumptions about $\beta$, $\sigma_y^2$
  - $\beta \sim N(0, \sigma_b^2 I_p)$
  - $\sigma_y^2 \sim IG(A, B)$
- Joint posterior of interest is

$$\pi(\beta, \sigma_y^2 | y, x, A, B, \sigma_b^2)$$

- Need full conditionals
  - $\pi(\beta | \sigma_y^2, y, x, A, B, \sigma_b^2)$
  - $\pi(\sigma_y^2 | \beta, y, x, A, B, \sigma_b^2)$

# Full conditionals

$$\beta \mid \sigma_y^2, y \sim \mathcal{N}(\beta_{\text{post}}, \Sigma_{\text{post}})$$

$$\Sigma_{\text{post}} = \left( \frac{1}{\sigma_y^2} X^\top X + \frac{1}{\sigma_\beta^2} I_p \right)^{-1}$$

$$\beta_{\text{post}} = \left( X^\top X + \frac{\sigma_y^2}{\sigma_\beta^2} I_p \right)^{-1} (X^\top y)$$

$$\sigma_y^2 \mid \beta, y \sim IG\left( A + n/2, B + \frac{1}{2} ||y - x\beta||^2 \right)$$

# Variations on Gibb's sampling

- **Blocked Gibb's sampling**: update blocks (or groups) of parameters jointly rather than one at a time
  - Helps improve mixing when variables are highly correlated
  - We are already doing this for Bayesian linear regression
- **Collapsed Gibb's sampling**: integrate/marginalize out some variables analytically before sampling
  - Reduces dimensionality and can improve convergence
- **Metropolis-within-Gibbs**: Uses Metropolis-Hastings steps for full conditionals that are not easily sampled from.
  - Commonly used in Bayesian hierarchical models

# Drawbacks of Gibbs and Random walk MH

- Tend to get stuck in small regions of the posterior for a long time
  - Especially for large number of parameters
  - I.E. slow mixing for correlated parameters
  - Potentially low effective sample size
- Requires closed-form full conditionals
- Random walk Metropolis-Hastings can be slow because it slowly explores the state space

# Hamiltonian Monte Carlo

*Though originating in physics, Hamiltonian dynamics can be applied to most problems with continuous state spaces by simply introducing fictitious "momentum" variables.*

*HMC alternates updates for momentum variables with Metropolis updates in which a new state is proposed by computing a trajectory according to Hamiltonian dynamics. A state proposed in this way can be distant from the current state but nevertheless have a high probability of acceptance.*

# Hamiltonian Monte Carlo

- The first step is to define a Hamiltonian function in terms of the probability distribution we wish to sample from.
- Parameters of interest are called “position” variables
- Also introduce auxiliary “momentum” variables

# Hamiltonian dynamics

*Hamiltonian dynamics has a physical interpretation that can provide useful intuitions. In two dimensions, we can visualize the dynamics as that of a frictionless puck that slides over a surface of varying height The state of this system consists of the position of the puck, given by a 2D vector $q$, and the momentum of the puck (its mass times its velocity), given by a 2D vector $p$. The potential energy, $U(q)$, of the puck is proportional to the height of the surface at its current position, and its kinetic energy, $K(p)$, is equal to $|p|^2/(2m)$, where $m$ is the mass of the puck. On a level part of the surface, the puck moves at a constant velocity, equal to $p/m$. If it encounters a rising slope, the puck's momentum allows it to continue, with its kinetic energy decreasing and its potential energy increasing, until the kinetic energy (and hence $p$) is zero, at which point it will slide back down (with kinetic energy increasing and potential energy decreasing).*

# Hamiltonian dynamics

$$H(q,p) = U(q) + K(p)$$

- U(q): potential energy
- K(p): kinetic energy
  - Usually defined as $p^T M^{-1} p/2$, where $M$ is a "mass matrix"

# Hamiltonian dynamics

The partial derivatives of the Hamiltonian determine how $q$ and $p$ change over time $t$, according to Hamilton's equations:

$$\frac{dq_i}{dt} = \frac{\partial H}{\partial p_i}$$

$$\frac{dp_i}{dt} = -\frac{\partial H}{\partial q_i}$$

# Hamiltonian dynamics in statistics

- **position** corresponds to parameters of interest
- **potential energy U(q)** is the (joint) log probability density of interest
- **kinetic energy K(p)** is usually defined as $p^T M^{-1} p/2$, where $M$ is a "mass matrix"
  - Corresponds to zero-mean Gaussian with covariance $M$

Can be used to construct MCMC updates.

# Gibbs sampler for Bayesian regression

I started implementing this myself… for lab we will go the rest of the way there and compare it to the results of `stan_glm` from the Bayesian Data Analysis lecture.

- My partial Gibb's sampler is in the file `gibbs_blr.R` and the lab file is `lab_gibbs.Rmd`.
- Compare effective sample size for your Gibb's sampler and for Stan's HMC algorithm
- Compare diagnostic plots as well

# Resources

- Original Gibb's sampling paper
- Book chapter on Hamiltonian MC
- Stan manual
- No U-Turn Sampler (NUTS)
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